Answer Key & Teacher Notes

Session 1 — Find the Percent of a Number · Session 2 — Compare Percents

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages: 8 problems, Find the Error, 2 test items. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Teaching Support Guide	Teacher page: lesson route, what to look for on each exit item, the next prompt, and a fresh check for after reteaching.
Explain Your Thinking	Optional language support for the Guided Notes "Model, Reason, Explain" page: draw it, talk it, write it.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank (English and Spanish), step frames on every problem and larger type (12 pt).

SESSION 1 · TEACHER NOTES

Find the Percent of a Number

TALK PROMPTS

- Why do you multiply by the percent written in hundredths instead of by the percent number itself?
- Before you multiply, how do you know 35% of 160 has to be less than 160?

PARTNER TASK

Music Playlist and Book Pages. One partner writes the percent as a decimal and the equation; the other multiplies and checks that the part is less than the whole. Switch roles with Pizza Orders: 35% of 160 pizzas (56).

WATCH FOR

- Multiplying by the percent number (30 × 70) instead of the decimal (0.30 × 70).
- Writing small percents with the wrong place value (6% as 0.6, 9% as 0.9).
- Dividing the whole by the percent (80 ÷ 25 for 25% of 80) instead of multiplying by it.

SESSION 2 · TEACHER NOTES

Compare Percents

TALK PROMPTS

- Why does turning each record into a percent make the comparison fair?
- Friday's club had more members (65 vs. 60). Why does that not decide which day a greater share came?

PARTNER TASK

Club Days. One partner finds the Monday and Friday percents (75%, 60%); the other explains which day was better and why counts alone do not decide. Switch roles with Recycling.

WATCH FOR

- Ranking records by the part alone (22 out of 40 "beats" 12 out of 15).
- Comparing the part to 100 instead of to its own whole (33 out of 55 called 33%).
- Missing a tie when two different fractions simplify to the same percent (34/40 and 51/60).

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.AT.4	1	2	1	1	2	2	2	3	3	1	1
S1 · Form B	6.AT.4	1	2	1	1	2	2	2	3	3	1	1
S2 · Form A	6.AT.4	1	2	1	1	2	2	2	3	3	1	1
S2 · Form B	6.AT.4	1	2	1	1	2	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

part: The piece of the whole that the **percent** tells about.

whole: The full amount. It comes right after the word "**of**."

equation: A math sentence with an **equal** sign.

- Write the percent as a **decimal**: 45% = 0.45 and 6% = 0.06.
- Multiply: part = percent × **whole**.
- If the percent is less than 100%, the part is **less** than the whole.

Example: The whole is **140** tickets. Each of the 20 parts is 5%. 45% = **0.45** 0.45 × 140 = **63** Answer: **63** student tickets

YOU TRY

1. a. 12 b. 20 c. 27

2. a. 0.15 × 640 = 96 students absent b. 640 − 96 = 544 students present

EXIT TICKET

1. a. 36 b. 30

2. 0.06 × 45 = $2.70

3. **C** (0.15)

PRACTICE FORM A

1. a. 20 b. 23 c. 21 d. 7 e. 44 f. 36

2. Milk 0.30 × 200 = 60; water 0.55 × 200 = 110; juice 100% − 85% = 15%, 0.15 × 200 = 30. Check: 60 + 110 + 30 = 200.

3. Bar: 20, 40, 60. 0.25 × 80 = 20 songs

4. 100 × 0.35 = 35 and 160 × 0.35 = 56. 56 pizzas

5. a. 0.65 × 320 = 208 pages b. 320 − 208 = 112 pages

6. No. Discount = 0.20 × 48 = $9.60. Sam used the percent number as dollars.

7. a. 0.40 × 180 = 72 peppers b. 100% − 75% = 25%; 0.25 × 180 = 45 seedlings

8. a. 0.65 × 1,200 = 780 tickets b. 780 × $18 = $14,040 c. 1,200 − 780 = 420 empty seats, which is 35% of the hall

9. 30% is 0.30, not 30. A part of $70 must be less than $70. 0.30 × 70 = $21.

10. **B** (30)

11. **C** (0.45)

Self-check word **DECIMAL**

PRACTICE FORM B

1. a. 16 b. 47 c. 18 d. 13 e. 28 f. 30

2. Milk 0.20 × 300 = 60; water 0.45 × 300 = 135; juice 100% − 65% = 35%, 0.35 × 300 = 105. Check: 60 + 135 + 105 = 300.

3. Bar: 15, 30, 45. 0.25 × 60 = 15 songs

4. 100 × 0.35 = 35 and 140 × 0.35 = 49. 49 pizzas

5. a. 0.85 × 240 = 204 pages b. 240 − 204 = 36 pages

6. No. Discount = 0.20 × 65 = $13. Ari used the percent number as dollars.

7. a. 0.45 × 160 = 72 peppers b. 100% − 75% = 25%; 0.25 × 160 = 40 seedlings

8. a. 0.85 × 800 = 680 tickets b. 680 × $15 = $10,200 c. 800 − 680 = 120 empty seats, which is 15% of the hall

9. 9% is 0.09, not 0.9 (0.9 is 90%). 0.09 × 200 = 18.

10. **B** (32)

11. **C** (0.15)

Self-check word **PERCENT**

GUIDED NOTES

part-to-whole ratio: Compares part of a group to the **whole** group.

common whole: The same total, like **100**, used for every group.

compare: To decide which amount is **greater** or less.

- A record (how many out of how many) can fool you when the **wholes** are different.
- Write each record as a **percent**, then compare the percents.

Example: Mia: 21 ÷ 28 = **0.75** = 75% Leo: 16 ÷ 20 = **0.80** = 80% Compare: 80% > 75%. Each small part of a bar is 5%. Answer: **Leo** has the better rate, even though Mia made more shots.

YOU TRY

1. a. 75% b. 72%

2. Ana 70%, Ben 68%. Ana has the better rate.

EXIT TICKET

1. a. 90% b. 35%

2. a. 15/20 = 75% b. 21/30 = 70% c. Team A

3. **B** (22 out of 25)

PRACTICE FORM A

1. a. 90% b. 80% c. 75% d. 84% e. 80% f. 60%

2. Dara 75%, Ivan 65%, Nora 80%. Nora is the best shooter, even though she did not take the most shots.

3. a. A 85%, B 85% b. tie

4. a. Monday 75%, Friday 60% b. Monday

5. Rico 9/12 = 3/4 = 75%; Tia 15/24 = 5/8 = 62.5%. Rico had the better rate.

6. No. 33/55 = 3/5 = 60/100 = 60%. Ben compared 33 to 100 instead of to 55.

7. Ella 63/70 = 90%; Omar 46/92 = 50%; 90 − 50 = 40 percentage points.

8. a. Class 1: 36/45 = 80% Class 2: 40/50 = 80% Class 3: 30/40 = 75% b. Classes 1 and 2 tie for first; Class 3 is third c. Class 1 also has 80% (36 of 45), so they tie. Class 2 recycled more items only because it had more items (50); a fair comparison uses percents, not counts.

9. Theo compared only the parts, not the parts to their wholes. 12/15 = 80% and 22/40 = 55%, so 12 out of 15 is the better score.

10. **C** (90%)

11. **A** (12 out of 15)

Self-check word **COMPARE**

PRACTICE FORM B

1. a. 85% b. 60% c. 25% d. 76% e. 55% f. 70%

2. Amir 33/44 = 3/4 = 75%, Bea 27/45 = 3/5 = 60%, Cole 34/40 = 85%. Cole is the best shooter.

3. a. C 45/50 = 90%, D 36/40 = 90% b. tie

4. a. Tuesday 36/48 = 75%, Thursday 44/55 = 80% b. Thursday

5. Jada 14/16 = 7/8 = 87.5%; Ken 18/20 = 90%. Ken had the better rate.

6. No. 28/40 = 7/10 = 70/100 = 70%. Mae compared 28 to 100 instead of to 40.

7. Lia 72/80 = 90%; Sam 33/60 = 55%; 90 − 55 = 35 percentage points.

8. a. Class 1: 24/30 = 80% Class 2: 42/60 = 70% Class 3: 35/50 = 70% b. Class 1 first; Classes 2 and 3 tie for second c. Class 2 had the most items (60), so 42 is only 70% of its total. Class 1 recycled a greater share (80%); a fair comparison uses percents, not counts.

9. Rae compared counts with different totals. 30/60 = 50% and 20/25 = 80%, so 20 out of 25 is the better record.

10. **C** (80%)

11. **A** (9 out of 10)

Self-check word **RECORDS**

SHEET 1 · FIND THE PERCENT OF A NUMBER

1. a. 9 b. 48 c. 21 d. 36 e. 45 f. 30
2. Missing: 72, 108, 144. 40% of 180 = 72 students.
3. a. 0.25 × 64 = $16 b. $64 − $16 = $48
4. 10% of 40 = 4 and 5% = 2, so 15% = 6 players (0.15 × 40 = 6).
5. They are equal: 0.3 × 90 = 27 and 0.9 × 30 = 27. Both equal 30 × 90 ÷ 100, so switching the numbers gives the same amount.

SHEET 2 · COMPARE PERCENTS

1. a. 70% b. 65% c. 75%
2. Ava 45%, Ben 48%, Cruz 44%. Ben has the best record.
3. Rosa: 34/40 = 85/100 = 85%. Kai: 42/50 = 84/100 = 84%. Rosa did better, even though Kai spelled more words.
4. A, C (75% and 72%; 17 out of 25 = 68%, 7 out of 10 = 70% exactly)
5. Equal rates come from multiplying, not adding the same amount. 18/24 = 3/4 = 75% and 24/30 = 4/5 = 80%, so the rates are not equal; 24 out of 30 is better.
6. **C** (33 out of 50)
7. **B** (A won 75%, B won 60%. A did better.)

SESSION 1 · FIND THE PERCENT OF A NUMBER

a. Bar: 36, 72, 108, 144, 180, 216, 252, 288, 324. 10% of 360 = 36 and 5% = 18, so 35% = 36 + 36 + 36 + 18 = 126 seats. b. part = 0.35 × 360 = 126 seats. c. Both ways take 35 out of every 100 parts of the same whole, 360: 0.35 = 0.10 + 0.10 + 0.10 + 0.05. 126 is less than half of 360 (180), so it makes sense.

Challenge 35% is less than half, so the part must be less than 180 (half of 360). 12,600 is more than all 360 seats. Sam multiplied by 35 instead of 0.35. 0.35 × 360 = 126 seats.

SESSION 2 · COMPARE PERCENTS

a. Kira: 36/60 = 6/10 = 60/100 = 60% (shade 6 of 10 parts). Dev: 32/40 = 8/10 = 80/100 = 80% (shade 8 of 10 parts). b. Dev. c. Kira made more free throws, but she also took more shots (60 vs. 40). Each percent compares the part to its own whole, and 80% > 60%.

Challenge 54/60 = 9/10 = 90/100 = 90%, and 9 out of 10 = 90%. The two records tie. 54 is bigger only because its whole (60) is bigger.

Score each task 0–3: 1 for a model that fits the numbers, 1 for correct math with units, 1 for an explanation that matches the model.